Response to the Reviewers

Paper ID 1343: *Real-Time Adaptive Cross-Media Recommendation Using Semantic Embeddings, Knowledge Graph Scoring, and Graph Collaborative Learning*

International Conference on Machine Learning and Data Engineering (ICMLDE 5.0)

Dear Scientific Committee and Reviewers,

We thank the reviewers for their careful reading and constructive comments. We have revised the manuscript and respond to every comment below, quoting each comment in italics and giving what we changed and where. The main changes are:

- **Larger evaluation** (Section 4 (p. 6), Section 5 (p. 7)): three Amazon Reviews 2023 datasets with up to 50,000 users, reporting Recall, NDCG, MAP, and MRR at $K \in \{10, 20\}$ (plus HitRate and Precision) with significance tests (Table 2 (p. 7)), and 17 prototype scenarios instead of 5 (Table 6 (p. 9)).
- **Full ablation**: all 16 combinations of the four signals (semantic, EMA, LightGCN, knowledge graph), each with weights selected on a validation split (Table 4 (p. 8)).
- **Baselines**: most-popular and single-signal rankings evaluated under the same protocol as the hybrid (Table 4 (p. 8)).
- **Configuration and latency**: complete experimental settings (Table 3 (p. 7)) and measured latency and throughput (Table 5 (p. 9)).
- **Method descriptions**: event weights, EMA initialization and negative updates, LightGCN training, and the knowledge-graph construction (Section 3 (p. 3)).
- **Related work**: Table 1 (p. 3) updated and Section 2 (p. 2) extended with 2025–2026 work.
- **Format and language**: Procedia Computer Science template, redrawn figures, shortened captions, and proofreading of the entire manuscript.

In the revised manuscript, the lines that address each comment and the new or redrawn tables and figures are highlighted in yellow; the font (Comment R1.2), the proofreading (Comment R1.6), and the template (Comment R1.7) apply to the entire manuscript and are not highlighted.

Numbering in the submitted and revised manuscripts

Section, table, figure, and equation numbers differ between the two versions, so each comment below lists its location in both. Page numbers of the submitted version are those of the PDF we submitted; page numbers of the revised version are those of the attached manuscript. A table or figure that appears under the same number in both versions may show different content, as noted.

	Submitted version	Revised version
Introduction	Section 1 (p. 2)	Section 1 (p. 1)
Related work	Section 2, "Discussion of Related Work and Novelty" (pp. 2–3)	Section 2, "Related Work and Novelty" (p. 2)
System overview	Sections 3.1 and 3.9 (pp. 4, 8)	Section 3.1 (p. 3)
Data and catalog	Section 3.2 (p. 4)	Section 3.2 (p. 3)
Semantic retrieval, creator and title matching	Sections 3.3 and 3.4 (p. 5)	Section 3.3 (p. 4)
EMA	Section 3.5 (p. 6)	Section 3.4 (p. 4)
LightGCN	Section 3.6 (p. 6)	Section 3.5 (p. 4)
Knowledge graph	Section 3.7 (p. 7)	Section 3.6 (p. 5)
Hybrid ranking	Section 3.8 (p. 8)	Section 3.7 (p. 6)
Experimental setup	— (new section)	Section 4 (p. 6)
Results	Section 4 (p. 9)	Section 5: baselines and ablation 5.1, latency 5.2, prototype scenarios 5.3 (p. 7)
Conclusion	Section 5 (p. 10)	Section 6 (p. 10)
Table 1	Comparison of prior model families and measures (p. 3)	Table 1, *updated*: comparison with recent graph, knowledge-aware, and cross-domain recommenders (p. 3)
Table 2	Module-wise input, output, and algorithm (p. 8)	Removed; the content is in Section 3 and Fig. 1
Table 2 (revised)	—	Table 2, benchmark graphs (p. 7)

	Submitted version	Revised version
Table 3	Prototype results with qualitative single-signal rows (p. 9)	Qualitative rows replaced by the quantitative Table 4 (p. 8); per-scenario results in Table 6 (p. 9)
Table 3 (revised)	—	LightGCN and EMA settings (p. 7)
Table 5 (revised)	—	Latency (p. 9)
Figure 1	Architecture (p. 4)	Fig. 1, redrawn (p. 3)
Figure 2	Knowledge graph (p. 7)	Fig. 2, redrawn (p. 5)
Figure 3	Result bar chart (p. 9)	Removed; its values are in Table 6
Equations	(1) catalog, (2) semantic score, (3) EMA update, (4) EMA score, (5)–(6) LightGCN, (7) BPR, (8) graph score, (9) paths, (10) KG score, (11) profile score, (12) hybrid	(1) catalog, (2) EMA update, (3) LightGCN, (4) BPR, (5) paths, (6) KG score, (7) profile score, (8) hybrid, (9) preference bonus (new); the semantic, EMA, and graph scores are now defined in the text

Reviewer 1

Comment R1.1. *"The Figure 1 caption is written twice."*

Response. We redrew Figure 1 without embedded text, so it now has a single caption, and updated it to the current data sources.

Submitted version: Fig. 1, p. 4.

Revised version: Fig. 1 (p. 3).

Comment R1.2. *"Different text fonts are used on pages 4 and 5."*

Response. We changed the font of the entire manuscript to Times, the template font, including the text inside both figures.

Submitted version: pp. 4–5 (Sections 3.1–3.4 and Fig. 1); the same typeface also appeared on later pages.

Revised version: Entire manuscript; Section 3 (pp. 3–6), Fig. 1 (p. 3), Fig. 2 (p. 5).

Comment R1.3. *"Check the caption of Figure 2."*

Response. We shortened the caption to one sentence and redrew Figure 2 so that its relation labels match the relations defined in the knowledge-graph subsection.

Submitted version: Fig. 2, p. 7 (Section 3.7).

Revised version: Fig. 2 (p. 5), Section 3.6 (p. 5).

Comment R1.4. *"Correct the captions of all tables."*

Response. All table captions now follow the template: one line above the table, with explanatory text moved to the body. Tables 1–3 of the submitted version correspond to revised Tables 1, 4, and 6 (the submitted Table 2 was removed); the new tables follow the same rule.

Submitted version: Tables 1–3 (pp. 3, 8, 9).

Revised version: Tables 1–6 (pp. 3–9).

Comment R1.5. *"Cite papers from 2025 onwards."*

Response. We added six 2025–2026 papers: LGKAT [18], LightKG [19], LLMCDSR [20], LLM4CDSR [21], LeCDSR [22], and KLLMRec [23]. They are discussed in Section 2 and compared in Table 1; the works [11, 16, 17] are also from 2025–2026.

Submitted version: Section 2 (pp. 2–3); References [1]–[20].

Revised version: Section 2 (p. 2); Table 1 (p. 3); References [18]–[23].

Comment R1.6. *"Spelling mistakes (e.g., in conclusion)."*

Response. We corrected the following five mistakes and proofread the entire manuscript:

(1) "papaer" → "paper" (Conclusion);

(2) "differnent" → "different" (Section 2);

(3) "retrival" → "retrieval" (Section 2);

(4) “choosen” → “chosen” (Section 2);

(5) “go show” → “to show” (Introduction).

The citation that printed as “[?]” in Section 2 (the SASRec reference) was also fixed.

Submitted version: “papaer”: Section 5, p. 10; “differnent”, “choosen”, and “[?]”: Section 2, p. 3; “retrival”: Section 2, p. 2; “go show”: Section 1, p. 2.

Revised version: Section 6 (p. 10); Section 2 (p. 2); Section 1 (p. 1); entire manuscript.

Comment R1.7. *“Ensure the paper is in the template.”*

Response. The manuscript now uses the Procedia Computer Science template without modifications: custom spacing, font, caption, and list settings were removed, and the running head, the corresponding-author footnote, and the reference style follow the template.

Submitted version: Entire manuscript.

Revised version: Entire manuscript.

Reviewer 2

Comment R2.1. *“The proposed integration of Sentence-BERT, Qdrant, EMA personalization, LightGCN, and knowledge-graph/profile scoring is interesting and clearly presented. However, the manuscript should strengthen the novelty discussion by providing a more detailed comparison with recent cross-domain and hybrid recommendation approaches.”*

Response. We updated Table 1 (it now compares LightGCN, KGAT, LGKAT, LightKG, LLMCDSR, LLM4CDSR, and this work on domains, text semantics, collaborative graph, knowledge graph, short-term adaptation, and deployment) and extended Section 2 with recent knowledge-aware graph recommenders [18, 19] and LLM-based cross-domain recommenders [20–23]. The novelty paragraph now states that the contribution is the integration of these signals over one catalog, with separate immediate (EMA) and periodic (LightGCN) personalization, together with the trade-offs of this design.

Submitted version: Section 2 (pp. 2–3); Table 1 (p. 3).

Revised version: Section 2 (p. 2); Table 1 (p. 3).

Comment R2.2. *“The experimental evaluation is limited to five prototype scenarios. Please provide a larger-scale evaluation using standard recommendation metrics such as Recall@K, NDCG@K, MAP, and MRR, as the current study itself identifies these as future evaluation requirements.”*

Response. We extended the evaluation in two ways.

(1) *Offline benchmark.* Three Amazon Reviews 2023 datasets [27]: Movies & TV (50,000 users, 52,213 items), Health & Household (50,000 users, 42,621 items), and Industrial & Scientific (33,959 users, 18,804 items). Each method ranks the full catalog for every test user. We report Recall, NDCG, MAP, and MRR at $K = 10$ and $K = 20$, together with HitRate and Precision, and paired Wilcoxon signed-rank tests.

(2) *Prototype scenarios.* We increased the scenarios from 5 to 17 (five person queries, six cross-media intents, and six queries in the health, industrial, and finance verticals), ran all of them through the deployed API, and added Precision, Recall, NDCG, MAP, and MRR at 10 to the earlier measures. The results of every scenario are in Table A below.

Submitted version: Section 4 (p. 9); Table 3 (p. 9); Fig. 3 (p. 9).

Revised version: Section 4 (p. 6), Table 2 (p. 7); Section 5.1 (p. 7), Table 4 (p. 8); Section 5.3 (p. 9), Table 6 (p. 9).

Table A. Results of the 17 prototype scenarios (PH: PersonHit, PA: ProfileAlign, KG and Prof.: mean KG and profile score of the top five; P, R, N, MAP, MRR: rank metrics at 10 against rule-based relevance judgments, defined in Section 5.3 (p. 9)).

Scenario	Age / domain	PH@5	PA@5	KG	Prof.	P@10	R@10	N@10	MAP@10	MRR@10	Content types in top 10
James Cameron	20	4/5	5/5	0.667	0.980	0.800	0.533	0.716	0.617	0.500	finance, movie, show
Taylor Swift	20	5/5	5/5	0.667	0.630	0.800	0.216	0.870	0.800	1.000	book, movie, music
Business analytics	30	–	5/5	0.650	1.000	0.900	0.016	0.927	0.866	1.000	book, finance
Fun animation adventure	8	–	5/5	0.667	1.000	1.000	0.002	1.000	1.000	1.000	movie
Space adventure with aliens	20	–	5/5	0.600	0.896	1.000	0.006	1.000	1.000	1.000	movie, show
Joint pain supplements	30, health	–	5/5	0.900	1.000	1.000	0.003	1.000	1.000	1.000	health
Lab safety gloves	30, industry	–	5/5	0.767	0.694	1.000	0.010	1.000	1.000	1.000	industrial
Personal budgeting	30, finance	–	5/5	0.700	0.938	1.000	0.011	1.000	1.000	1.000	finance
Christopher Nolan	30	5/5	5/5	0.667	0.640	0.800	0.533	0.848	0.733	1.000	movie
Stephen King	30	4/5	5/5	0.667	0.834	0.900	0.125	0.890	0.790	1.000	book, show
Coldplay	20	4/5	5/5	0.333	0.502	0.900	0.237	0.780	0.707	0.500	movie, music
Magic and dragons	20	–	5/5	1.000	0.958	1.000	0.007	1.000	1.000	1.000	book, movie, show
Romantic drama about love	30	–	5/5	0.550	0.992	1.000	0.002	1.000	1.000	1.000	movie, show
Action with superheroes	30	–	5/5	0.650	0.970	1.000	0.003	1.000	1.000	1.000	movie
Vitamin D and immunity	30, health	–	5/5	1.000	0.938	1.000	0.003	1.000	1.000	1.000	health
Torque wrench and fasteners	30, industry	–	5/5	0.760	0.854	1.000	0.012	1.000	1.000	1.000	industrial
Retirement investing	30, finance	–	5/5	0.600	0.758	1.000	0.014	1.000	1.000	1.000	finance
Mean		0.88	1.00	0.697	0.858	0.947	0.102	0.943	0.913	0.941	8/17 cross-domain

Comment R2.3. *"A detailed ablation study is recommended to quantify the individual contribution of semantic retrieval, EMA personalization, LightGCN, and KG/profile scoring. This would help demonstrate whether the improvement is actually attributable to the proposed hybrid architecture. The final score currently combines these components using fixed weights."*

Response. We added a full ablation of the four signals (semantic S, EMA E, LightGCN G, knowledge graph K): all 16 on/off combinations, evaluated on the three datasets with Recall, NDCG, MAP, and MRR (Table 4). The empty combination is most-popular ranking and S+E+G+K is the proposed hybrid. The weights are no longer fixed: each combination has its own weights, selected by grid search on a separate validation split, and the hand-set weights are listed as a reference row. The declared-profile score has no counterpart in public interaction logs, so it is ablated in the prototype scenarios (see R3.6). Table B gives NDCG@20 and Recall@20 of the 16 combinations: on Movies & TV and Industrial & Scientific the four-signal hybrid is the best of all 16 on every measure, and on Health & Household the knowledge-graph signal alone is best.

Submitted version: Section 3.8, Eq. (12) (p. 8); Section 4 (p. 9).

Revised version: Section 4 (p. 6) (baselines and weight selection); Section 5.1 (p. 7); Table 4 (p. 8); Section 5.3 (p. 9).

Table B. NDCG@20 (N) and Recall@20 (R) of all 16 combinations of the four signals; the best value in each column is in bold, and ° marks a combination whose NDCG@20 does not differ significantly from S+E+G+K ($p \geq 0.05$). The full table with MAP and MRR is Table 4.

Signals	Movies & TV		Health & Household		Industrial & Scientific	
	R@20	N@20	R@20	N@20	R@20	N@20
None (popularity)	0.0197	0.0097	0.0237	0.0115	0.0282	0.0147
S	0.0307	0.0187	0.0166	0.0096	0.0379	0.0223
E	0.0233	0.0128	0.0104	0.0052	0.0254	0.0135
G	0.0257	0.0136	0.0116	0.0054	0.0242	0.0130
K	0.0093	0.0045	**0.0253**	**0.0144**	0.0364	0.0210
S+E	0.0309	0.0188	0.0169	0.0097	0.0380	0.0225
S+G	0.0359	0.0222	0.0212	0.0120	0.0468	0.0267
S+K	0.0313	0.0193	0.0170	0.0097	0.0387	0.0226
E+G	0.0307	0.0174	0.0187	0.0092	0.0377	0.0205
E+K	0.0247	0.0141	0.0144	0.0072	0.0314	0.0162
G+K	0.0265	0.0141	0.0200	0.0097	0.0386	0.0207
S+E+G	0.0362	0.0224	0.0215	0.0122	0.0490	0.0277
S+E+K	0.0318	0.0197	0.0171	0.0098	0.0389	0.0227
S+G+K	0.0366	0.0226	0.0224	0.0126°	0.0484	0.0275
E+G+K	0.0314	0.0177	0.0232	0.0115	0.0410	0.0225
S+E+G+K	**0.0368**	**0.0227**	0.0223	0.0126	**0.0495**	**0.0283**
Hand-set weights	0.0344	0.0216	0.0208	0.0122	0.0455	0.0266

Comment R2.4. *"The baseline comparison should be strengthened. Table 3 mainly provides qualitative descriptions of semantic-only, graph-only, and KG/profile-only systems. Implementing these baselines under the same experimental conditions and reporting comparable quantitative results would provide stronger evidence for the effectiveness of the proposed approach."*

Response. *Quantitative:* the most-popular, semantic-only, EMA-only, LightGCN-only, and KG-only baselines are implemented in the same evaluation code as the hybrid, with the same chronological split, the same full-catalog candidate

set, and the same metrics; their results are the first rows of Table 4 and are compared with the hybrid in Section 5.1 (p. 7). *Qualitative:* Section 4 now describes what each baseline captures and what it misses, and Section 5.1 discusses how each behaves on each dataset. The qualitative rows of the submitted Table 3 were removed.

Submitted version: Table 3 (p. 9).

Revised version: Section 4 (p. 6) (baselines); Section 5.1 (p. 7); Table 4 (p. 8).

Comment R2.5. *"The reproducibility and real-time claims should be strengthened. Please provide complete experimental configurations, including data splitting, LightGCN training parameters, EMA settings, hybrid-score weight selection, and detailed latency/throughput measurements. The manuscript currently acknowledges that production latency benchmarking and larger-scale evaluation remain future work."*

Response. Section 4 now specifies the data preparation (rating threshold, k-core filtering, user sampling), the chronological split (last two positives for testing, the previous one for validation), the candidate set, the offline counterpart of each signal, the metrics, and the weight-selection grid. Table 3 lists all LightGCN and EMA settings, and Section 4 gives the training time. Section 5.2 and Table 5 report latency and throughput measured on 700 requests to the deployed endpoint over the full 120,747-item catalog: median 363 ms (p95 455 ms, p99 507 ms), EMA update 0.05 ms, and 2.7–3.3 requests per second per process. The conclusion no longer lists these evaluations as future work.

Submitted version: Section 4 (p. 9); Section 5 (p. 10).

Revised version: Section 4 (p. 6); Table 2 (p. 7); Table 3 (p. 7); Section 5.2 (p. 8); Table 5 (p. 9); Section 6 (p. 10).

Reviewer 3

Comment R3.1. *"Specify LightGCN's training settings, including layers, embedding size, negative sampling, and stopping criteria."*

Response. The settings are now stated in the text and in Table 3: 64-dimensional embeddings, 3 propagation layers (averaged), one uniformly sampled negative per positive pair (resampled if observed), Adam with learning rate 10^{-3} and L_2 weight 10^{-4}, batches of 4,096 pairs, and a fixed 60 epochs without early stopping, so that validation and test data do not influence training.

Submitted version: Section 3.6, Eqs. (5)–(7) (p. 6).

Revised version: Section 3.5 (p. 4), Eq. (3) (p. 4), Eq. (4) (p. 5); Table 3 (p. 7).

Comment R3.2. *"Clarify how positive interactions are selected and how different event types are weighted."*

Response. Event weights are now listed: view 0.4, click 0.6, like and bookmark 1.0, complete 1.2, skip -1.0, and an explicit rating r maps to $(r-3)/2$. An interaction is positive, and becomes a LightGCN edge, when its weight is positive; repeated events are merged into one edge, and skips and low ratings are excluded. For the Amazon benchmark, ratings of four or five stars are positive.

Submitted version: Section 3.5 (p. 6); Section 3.6 (p. 6).

Revised version: Section 3.4 (p. 4); Section 3.5 (p. 4); Table 3 (p. 7).

Comment R3.3. *"Explain how the EMA profile is initialized and how skips or other negative events update it."*

Response. The profile is empty until a user's first non-neutral event, which sets it to the item embedding (or its negation for a negative event); users without events receive no EMA score. Each event with weight w moves the profile by the step $\alpha \min(|w|, 1)$ with $\alpha = 0.25$, toward the item embedding for positive events and toward its negation for skips, and the profile is then renormalized to unit length (Eq. (2) (p. 4)).

Submitted version: Section 3.5, Eqs. (3)–(4) (p. 6).

Revised version: Section 3.4 (p. 4), Eq. (2) (p. 4).

Comment R3.4. *"Provide a precise procedure for constructing KG entities, relations, and user-item paths."*

Response. The knowledge-graph subsection now gives a three-step procedure: (i) entities (items typed by content type; creator, genre, and term entities from the catalog text; user and profile nodes), (ii) relations (directed_by, written_by, performed_by, belongs_to_genre, liked, bookmarked, completed, and the profile expansions with examples), and (iii) length-two user–entity–item paths, which are counted by set intersection in the KG and profile scores. Figure 2 was redrawn to match.

Submitted version: Section 3.7, Eqs. (9)–(11), Fig. 2 (p. 7).

Revised version: Section 3.6 (p. 5); Eq. (5) (p. 5), Eq. (6) (p. 5), Eq. (7) (p. 6); Fig. 2 (p. 5).

Comment R3.5. *"Reconcile the stated cross-domain output rate with the scenario results and define the calculation."*

Response. The cross-domain rate is now defined as the fraction of scenarios whose top ten results contain at least two content types, and Table A lists the content types of every scenario so that it can be checked directly. All scenarios were re-run through the deployed recommendation API and extended to 17. The rate is 8/17 overall and 8/11 for the scenarios without a domain filter (6/17 and 6/11 if movies and TV shows count as one medium); the six domain-scoped scenarios stay within their declared domain by design.

Submitted version: Section 4 and Table 3 (p. 9); abstract, p. 1 (cross-domain output "in 60% of tested scenarios").

Revised version: Section 5.3 (p. 9); Table 6 (p. 9).

Comment R3.6. *"Include ablation results to show the individual effects of semantic, EMA, graph, and KG/profile signals."*

Response. The offline ablation is the experiment described in R2.3 (Table 4 and Table B): the effect of each signal can be read from the 16 combinations on three datasets. In addition, the knowledge-graph and profile signals were ablated in the 17 prototype scenarios, run through the deployed system with each weight set to zero. Without the profile score, mean ProfileAlign@5 falls from 1.00 to 0.96, the mean profile score of the top five from 0.858 to 0.795, and NDCG@10 from 0.943 to 0.916, and on average 0.71 of the top five items change (most strongly for the pop-music user, from 5/5 to 2/5 profile-aligned items). Without the knowledge-graph score, the mean KG score of the top five falls from 0.697 to 0.664 and NDCG@10 from 0.943 to 0.931, with 0.47 of the top five items changed.

Submitted version: Section 4 (p. 9).

Revised version: Table 4 (p. 8); Section 5.1 (p. 7); Section 5.3 (p. 9).

Editorial requirements

- **Language:** the manuscript has been proofread and edited throughout.
- **Figures:** both figures are original and were drawn by the authors; no figure is taken from published work.
- **Template:** the manuscript follows the Procedia Computer Science template without modifications (see Comment R1.7).
- **Similarity:** the similarity report of the revised manuscript is uploaded as a supplementary file.

Additional updates

We also updated the following points in the manuscript:

- the authors of reference [12] (LLMRec), in the References;
- the graph score, now given as the cosine similarity of normalized embeddings, as implemented (Section 3.5 (p. 4));
- the data sources, now matching the current catalog of seven content types and 120,747 items, with the Last.fm API [26] in place of MusicBrainz (Section 3.2 (p. 3), Fig. 1 (p. 3));
- the ranking pipeline, which now also describes creator and title matching and the short-term preference bonus, so that Eq. (8) (p. 6) matches the implementation (Section 3.3 (p. 4), Section 3.7 (p. 6));
- the viewer-age and domain eligibility filter applied before ranking (Section 3.1 (p. 3)).

We hope that the revised manuscript addresses the reviewers' concerns, and we thank them again for their valuable comments.

Sincerely,
Sanjay A. K., Eswara Raj M., Aditya Vijay, Himaneesh Reddy Jangalapalli, Bagavathi Sivakumar P., Krishna Priya G.